

COGNIFY — STUDYMATE AI

Comprehensive Daily Project Execution Tree & Technical Deliverables
Summary

STATUS: **PRODUCTION READY**
VERSION: 1.2.0 (Stable)
DATE: September 18, 2026
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Core Pillars 4 Areas	Key Features Built 18 Completed	AI Engine Google Gemini	Status Live & Ready
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1. Summary of Everything Done Today (Simple Tree View)

Here is a simple, plain-English breakdown of all the features, design upgrades, and cloud deployment steps completed for Cognify today.

COGNIFY (AI STUDY ASSISTANT) – WHAT WE BUILT TODAY ■■■■ [1.0] SMART AI & STUDY CAPABILITIES ■■■■ Connected latest Google Gemini AI for smart, rapid study answers ■■■■ Added conversation memory (it remembers your previous questions) ■■■■ Built PDF Document Reader (upload lecture notes and ask questions on them) ■■■■ Added safety fallbacks so the app never crashes on AI rate limits ■■■■ Added "Clear Chat" button to wipe memory and start fresh ■■■■ [2.0] DESIGN, THEME & USER EXPERIENCE ■■■■ Rebranded to "Cognify" with a modern glowing synapse logo ■■■■ Upgraded color scheme from purple to modern Electric Blue & Sky Cyan ■■■■ Added Light Mode and Dark Mode switch (remembers your choice) ■■■■ Fixed bug where chat answers were hiding behind the search bar ■■■■ Made chat automatically scroll down when new answers appear ■■■■ Added Text-to-Speech button (reads AI answers out loud to you) ■■■■ Added Voice Dictation (speak into your microphone instead of typing) ■■■■ Added One-Click Copy buttons for quick note-taking and code ■■■■ [3.0] PHONE & MULTI-DEVICE ACCESS ■■■■ Made the site openable on any phone or tablet on the same Wi-Fi ■■■■ Fixed all website links so they work on phones without errors ■■■■ Verified Windows Firewall permissions so other devices can connect ■■■■ Created an instant public web link to use the site on mobile data anywhere ■■■■ [4.0] 24/7 CLOUD HOSTING & AUTOMATION ■■■■ Protected secret API keys so private credentials are never leaked ■■■■ Uploaded all project code to your GitHub (kaifhussain128-dev/cognify) ■■■■ Created a 1-click deploy script (deploy.bat) for instant future updates ■■■■ Set up Render cloud hosting so the website stays online 24/7 forever
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2. Plain-English Breakdown of Every Feature

Here is what each part of the project does, explained in straightforward, non-technical language.

[1.0] Smart AI & Study Features

COMPLETED

ID	Subtask / Feature	Artifact / Path	Implementation Details & Impact
1.1	Google Gemini AI Brain	backend/main.py	Connected the newest Google Gemini model to answer study questions accurately with explanations, bullet points, and code.
1.2	Continuous Memory	Chat System	The AI now remembers what you asked earlier in the conversation, allowing you to ask follow-up questions naturally.
1.3	PDF Document Reader	backend/pdf_processor.py	You can upload study slides, assignments, or textbook chapters (PDFs) and ask questions directly about the file.
1.4	Fresh Start Button	POST /clear-chat	Added a button that cleans the conversation memory and removes the attached PDF whenever you want to begin a new study topic.

[2.0] Design, Theme & User Experience

COMPLETED

ID	Subtask / Feature	Artifact / Path	Implementation Details & Impact
2.1	New Cognify Brand & Logo	Header & Logo	Replaced the generic title with 'Cognify' and designed a sleek glowing brain synapse logo in vibrant blue and cyan.
2.2	Electric Blue Theme	Colors & Styles	Replaced the dark purple theme with a clean, high-tech electric blue and sky cyan palette with soft ambient lighting.
2.3	Fixed Search Bar Overlap	Chat Window	Fixed the bug where answers got cut off behind the search bar. Now answers scroll comfortably in their own clear view.
2.4	Light & Dark Mode Switch	Theme Toggle	Users can toggle between a bright, clean Light Mode and an eye-friendly Dark Mode. The website remembers your choice automatically.
2.5	Listen (Text-to-Speech)	Audio Player	Added a button to have the AI read its answers out loud to you, making it easy to listen to study explanations.
2.6	Voice Typing (Dictation)	Microphone Button	You can click the microphone icon and speak your question instead of typing it out on your keyboard.
2.7	1-Click Copy Buttons	Message Cards	Every answer has a copy button so you can instantly paste notes, summaries, or code into your documents.

[3.0] Phone & Multi-Device Access

COMPLETED

ID	Subtask / Feature	Artifact / Path	Implementation Details & Impact
3.1	Home Wi-Fi Access	192.168.29.77:8000	Configured the server so you can open the app on your iPhone, Android phone, or iPad as long as you are on the same Wi-Fi.
3.2	Universal Device Links	frontend/index.html	Updated all backend connections so they work automatically from any phone or computer without setup errors.
3.3	Anywhere Public Link	Internet Tunnel	Created an instant public web link so you can share and test the website on cellular data or outside your home.

[4.0] 24/7 Cloud Hosting & Automation

● COMPLETED

ID	Subtask / Feature	Artifact / Path	Implementation Details & Impact
4.1	Safe API Key Storage	.gitignore & .env	Configured strict security rules so your private Google API key remains protected on your computer and is never leaked publicly.
4.2	Code Uploaded to GitHub	kaifhussain128-dev/cognify	Installed GitHub tools, linked your account, and uploaded all project code to your personal GitHub repository.
4.3	1-Click Deploy Tool	deploy.bat	Created a helper file (deploy.bat) so whenever you make changes in the future, 1 click sends them to GitHub and updates your live site.
4.4	Render 24/7 Cloud Host	render.yaml	Configured cloud hosting settings so Cognify can run on Render's free tier 24 hours a day without your PC staying on.

3. Final Deliverables & Repository Metadata

Resource	Endpoint / Target URI	Status / Verification
GitHub Repository	https://github.com/kaifhussain128-dev/cognify	Active / Up-to-date on main
Cloud Platform	https://dashboard.render.com/select-repo?type=web	render.yaml blueprint linked
Local Network (LAN)	http://192.168.29.77:8000	Accessible via Wi-Fi
Public Tunnel	https://heavy-seals-argue.loca.lt	External HTTPS tunnel ready
Automation Tool	Cognify/deploy.bat	1-Click automated deployer